

FOSSILS



THE REMAINS AND TRACES of past life forms are called fossils. All living organisms are potential fossils, but only a few are preserved. The most common fossils are those of hard parts of animals and plants. Only rarely is soft tissue fossilized. Sometimes, trace fossils, such as footprints, are found. The study of fossils, called palaeontology, is crucial to our understanding of life.

How a fossil is formed

In order for something to fossilize, it must be buried quickly by sediment, such as sand or mud, before it decomposes. Fossils form in a variety of ways, depending on the animal or plant lived, and the conditions after it was buried.



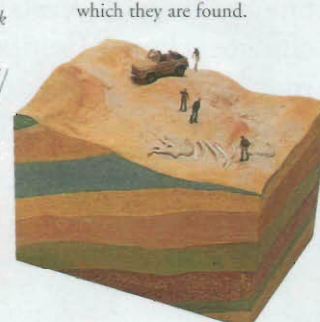
1 After death, the soft parts of a *Triceratops* decompose quickly, leaving just the hard skeleton and horns.



2 Through time, the bones are buried under thick layers of sediment and harden to form fossils.



3 The layers of sediment turn to rock. They may be pushed up or folded to form mountains.



Fossil dating

Some fossils, such as graptolites and conodonts, evolved and became extinct over geologically short periods of time. This makes them useful for dating the rocks in which they are found.



Graptolite



Conodont

Studying fossils

The study of the evolution of environments and natural communities is an important part of palaeontology. This limestone contains fossils of different animals, such as trilobites and corals. It shows a community that existed on the seafloor more than 400 million years ago.



Limestone from Much Wenlock, England

Coral

Fossil corals are common because they have a hard skeleton. Soft-bodied animals, such as sea anemones and jellyfish, would have lived in the same community, but are unlikely to have fossilized.



Trilobite

Trilobites were arthropods that lived in the sea. They shed their shells regularly, as modern arthropods do, and these shells are often found as fossils. They are divided into three distinct parts, or lobes, hence the name "trilobite". A few fossils of soft parts have been found.



Amber

The fossilized resin of trees, called amber, often contains trapped insects and other small animals and plants. The trapped fossils are often preserved with much detail.



Volcanic ash

This child's body was buried by volcanic ash at Pompeii, Italy, in AD 79. Ash turns to rock quickly. A buried animal or plant may rot away to leave a hollow, which, if filled with plaster, forms a cast.

Trees

Fossilized trees, such as these *Lepidodendron* trunks and roots, can be preserved as internal moulds of the bark. The inside rots away and is replaced by sand.



Petrification

These monkey-puzzle cones have been turned to stone, or petrified. This happened when silica-rich waters crystallized within the cells of the cones.



Concretions

Hard lumps, or concretions, are often formed around fossils in sediments. This concretion reveals the fossil shell and mould of a clam.

Types of fossil

Fossils range from microscopic plants and animals to the huge bones of dinosaurs. They can be almost unchanged from the original or replaced by minerals.

Bones

Vertebrate fossils are made of many parts and are usually found as single pieces. If conditions are right, a skeleton can be preserved whole, as in this *Diplomystus*, an ancestor of the modern herring.



Females were larger than males.

Ammonites

These molluscs were abundant in the seas of the Mesozoic Era. Their shells were made of the mineral aragonite and were often replaced by other minerals during fossilization.



Jurassic ammonite

Georges Cuvier

A French zoologist, Georges Cuvier (1769–1832) realized that the parts of the body were interrelated. For example, an animal that has hooves is a herbivore, and must have herbivore's teeth. He identified a fossil as a marsupial from a jaw.



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